

Position Extraction of Ultralow-Speed Gimbal Servo System With Linear Hall Sensors

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Abstract—This article explores a novel composite signal extraction method to extract high-precision angular position from the signals detected by Hall sensors in the gimbal servo system of single-gimbal magnetically suspended control moment gyro (SGMSCMG). Due to magnetic field distortion and other factors, the detected signals involve lots of harmonics and unequal amplitude, which decrease the position resolution accuracy significantly. In the proposed method, normalization is first performed to obtain signals with equal amplitude. Then, on the basis of offline analysis, the signal reconstructed model is established to suppress the most harmonics. Finally, the errors of the models and unmodeled noise are restrained by applying the discrete extended state observer (ESO). Simulated results show that the influence of harmonics and unequal amplitude on the angular position can be almost attenuated by the proposed method. Besides, plentiful experiments on an SGMSCMG demonstrate the validity of the proposed method, and the desired angular position of the ultralow speed servo system can be extracted effectively.

Index Terms—Extended state observer (ESO), linear Hall sensor, single-gimbal magnetically suspended control moment gyro (SGMSCMG), signal extraction method (SEM), ultralow speed.

I. INTRODUCTION

THE permanent magnet synchronous motors (PMSMs) have been widely used in a number of practical fields, and the gimbal servo motor in single-gimbal magnetically suspended control moment gyro (SGMSCMG) is one of the applications. SGMSCMG is a kind of important actuator for attitude adjustment with the advantages of a good dynamic performance, high control precision, and long service life [1]. As shown in Fig. 1, SGMSCMG is mainly composed of a high-speed rotor system and an ultralow-speed gimbal servo system. As the actuator, it is mainly applied in space, and considering the launch cost,

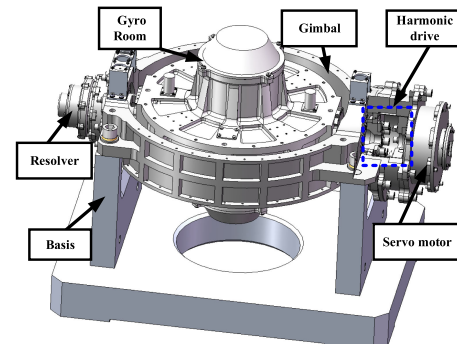


Fig. 1. Composition of SGMSCMG.

the control moment gyro has strict restrictions on its weight and volume. Therefore, the combination of every components in SGMSCMG also needs to be accurately calculated at the design stage. The output torque is generated by the gimbal changing the direction of high-speed rotor angular momentum that is $M = H_s \times \omega$ [2], where M indicates the output torque of CMG, the angular momentum H_s is provided by the high-speed rotor system, which often works at the state of constant and high speed, and the angular speed ω is provided by the gimbal servo motor system, which often works at the state of time-varying and low speed [3], [4]. Thus, the speed accuracy of the gimbal servo system directly affects the output torque accuracy of the SGMSCMG when the angular momentum is stable. In the control scheme of the gimbal servo system, accurate angular position is extremely indispensable, and there are two most widely applied typical sensors for position measurement, the encoder [5] and the resolver [6]. The optical incremental encoder exhibits outstanding performance in resolution. However, it is high priced, and the cost increases with the required stroke due to an additional scale. The sensitivity to contamination and environment limits its industrial applicability [7]. In the case of the resolver, the durability and the reliability are achievable by the sturdy hardware, and the position accuracy is high [8], nevertheless, its considerable size, weight, and cost remain an issue. In view of small size, light weight, and low cost, Hall sensors have been used by many investigators for rotor position sensing [9], [10].

On account of the size and weight limits to SGMSCMG, linear Hall sensors with small size, light weight, and low cost are appropriate for the position detection of PMSM in the gimbal servo system [11], [12]. The position information in the form of angle (radian) can be acquired through arctan2 calculation

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if sinusoidal and cosine signals measured by two linear Hall sensors with 90 electrical degrees apart are ideal [13], [14]. However, the detected signals can be easily affected by many factors such as the performance of the permanent magnets, and linear Hall sensor temperature drift. These factors can distort the magnetic field severely, and the signals contain lots of harmonics and unequal amplitude [15]. Many scholars have conducted research about the method to abstract accurate position information from the “dirty” signals. To make the raw detected signals ideal sinusoidal or cosine waveforms, nonideal components including the sensor offset, the scale difference between sensors, and the unwanted phase shift due to an alignment error have been compensated by adequate signal conditioning [13],[16]. The differential installation method of Hall sensors is very common adopted to mainly eliminate the unequal amplitude and dc component. Four Hall sensors are adopted to get the differential signals to remove dc component in the signals, and the results prove its effectiveness [17], [18]. As for the influence of the other factors, there are many different methods to obtain accurate position in different system. In the linear motor system, Bingran puts forward a fuzzy-neural offline modeling method to compensate the position errors introduced by Hall sensors, and the experimental results show the effectiveness of the method [19]. Jonghwa proposes the application of fast Fourier transformation and fixed-point interpolation to improve the resolution accuracy of the linear Hall sensors [20]. In the rotating motor system, in order to decrease the main disturbance (harmonics) of the raw detected signals, Song analyzes the interference in the signals and raises a same frequency extractor (SFE) to obtain the rotor fundamental frequency, and this method can be effectively applied in medium and high-speed motors [21]. The orthogonal PLL method is employed in [22]–[24] to attenuate the harmonics, which can be effective also in the medium and high-speed motor system. The state augmentation approach based on Kalman filter (KF) is applied by Leyang to remove third and fifth harmonics in the low-speed system [25], and this method is more suitable for signals with only low-order harmonics [26]–[28]. As for the signals with lots of harmonics in the low-speed system, the cascaded Kalman filter with the phase compensation and adaptive parameter adjustment method is present, and experimental results demonstrate the effectiveness, especially at the constant speed stage [29]. Besides, the state augmentation approach method [30], [31] can also be applied to handle the signals with much harmonics, but it will increase the computational effort which make the engineering application difficult.

The gimbal servo system in SGMSCMG often works at ultralow and time-varying speed in which the fundamental frequency of the motor is almost impossible to be acquired, and this is different from the previous work and is almost the hardest point. Meanwhile, both the dynamic and static characteristics need to be maintained good enough to provide accurate angular position in real time for the high-precision servo control [29],[32]. In view of the special case, the abovementioned methods are not effective in the gimbal servo system but more suitable for medium and high-speed systems, such as the SFE method. As for the KF method, which can works at low speed, it

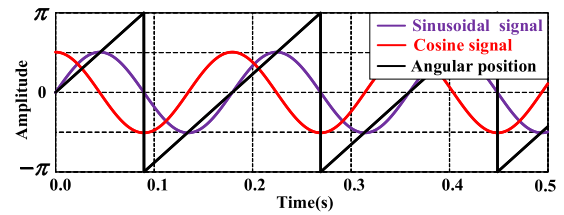


Fig. 2. Resolution principle of the Hall sensor's position.

is more suitable for a small number of harmonics, and the effect is better at a uniform speed. Consequently, in order to extract desired position information from the raw mixed signals in the gimbal servo system, a novel composite signal extraction method (SEM) based on signal normalization, signal reconstruction, and discrete extended state observer (ESO) is put forward in this article. The SEM is demonstrated by the simulation and experimental results with harmonics and other interference being suppressed, and a precise angular position for the gimbal servo system in SGMSCMG is provided. The contributions of this article are as follows. The accurate angular position information can be acquired without knowing the fundamental frequency information in advance in the low-speed system. Besides, the proposed method can also be applied in the medium speed system. Finally, the simple and practical process is conducive to the engineering application.

The rest of this article is organized as follows. The influencing factors and the description of the studied problem in this article are illustrated in Section II. In Section III, the details of the composite SEM are analyzed. Based on the analysis in Section III, the simulation can be found in Section IV. In Section V, the experimental setup is introduced, and plentiful experiments are presented to validate the proposed solutions. Finally, Section VI concludes this article.

II. CHARACTERISTICS OF THE RAW SIGNALS

A. Resolution Principle of Linear Hall Sensors

The working principle of the linear Hall sensor is to measure the magnetic field of the air gap in the motor [33], and the position information in the form of angle (radian) can be obtained through atan2 calculation when the sinusoidal and cosine signals measured by two linear Hall sensors with 90° (electrical degrees) apart are ideal [34]

$$\begin{cases} H_s = \sin(\theta) \\ H_c = \cos(\theta) \\ \theta = \text{atan2}(H_s, H_c) \end{cases} \quad (1)$$

where H_s and H_c are the output signals of the two Hall sensors, respectively, θ indicates the ideal angular position of the motor with period of 2π [35]. Fig. 2 depicts the resolution principle of the Hall sensor's position, which is consistent with the principle given by (1).

From (1) and Fig. 2, we can get that when the two signals are ideal, the obtained position accuracy is also ideal.

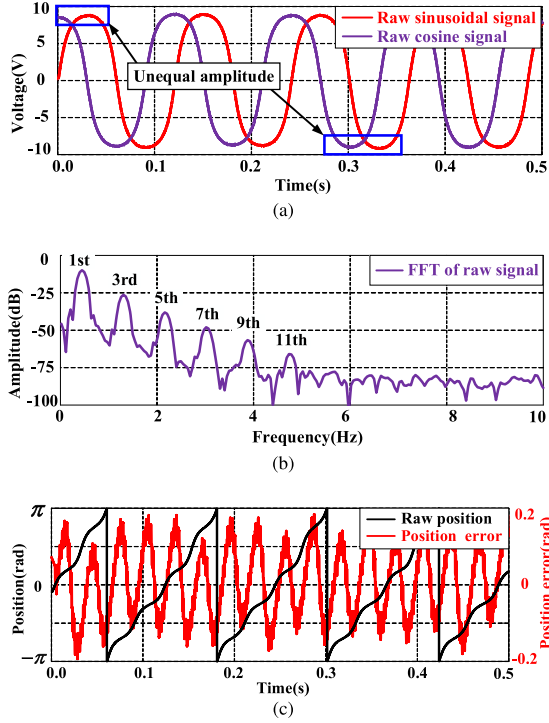


Fig. 3. Raw experimental information, (a) experimental sinusoidal and cosine signals, (b) its FFT, (c) raw position and position errors.

B. Description of the Studied Problem

From Section II-A, we know that the ideal position information mainly comes from the ideal sinusoidal and cosine signals. However, the magnetic field of the air gap can be easily distorted by many factors such as the inconsistent magnetization inhomogeneity of the magnetic steel, the harmonics in air gap magnetic field, nonideal antielectromotive force. Besides, the installation error of the Hall sensor, differential amplifier with zero drift, etc., can also cause a serious deterioration in the quality of the raw signals [15].

According to existing research [15], [21], it can be known that the aforementioned factors will cause the two signals to have different amplitude, nonorthogonal, dc components, lots of harmonics, and other problems, which will eventually result in the poor position accuracy. As for the problem of nonorthogonal and dc components, the influence can be reduced by adjusting the number and installation of the Hall sensors [29], and finally, two signals containing harmonic components and slightly unequal amplitude will be obtained. Fig. 3(a) shows the raw experimental conditioned signals, which are obtained by the commercial linear Hall sensors (THS130), and higher level devices will be employed in actual applications. Fig. 3(b) exhibits the FFT, the raw position, and the position errors are also given in Fig. 3(c).

It can be seen clearly from Fig. 3(a) that there are some differences in the amplitude of the two signals, which can result in position error [15]. From Fig. 3(b), we know that the harmonic components of the raw signals are mainly odd harmonics, and the raw angular position with the maximum error of about 0.18 rad in Fig. 3(c) is extremely poor, which can obviously not be adopted for precise servo control in the gimbal servo system.

Therefore, it is necessary to study how to extract the desired position information from the abovementioned signals with harmonics and unequal amplitude in the gimbal servo system of SGMSCMG, and the method is investigated in this article.

III. COMPOSITE SEM

In this section, the proposed novel composite SEM for the special issues mentioned in Section II is analyzed, which involves the signal normalization, signal reconstruction, and the discrete ESO method.

A. Signal Normalization

The two raw detected signals have been exhibited in Fig. 3, it can be found that the raw signals mainly contain the third-, fifth-, seventh-, and ninth-order harmonics, and the amplitude is not equal every period. According to the abovementioned facts, the two raw signals can be expressed as

$$\begin{cases} H_s(k) = \alpha \{ \sin[\theta(k)] + A_i \sin[i\theta(k)] \} \\ H_c(k) = \beta \{ \cos[\theta(k)] + B_i \cos[i\theta(k)] \} \end{cases} \quad (2)$$

where A_i and B_i ($i = 3, 5, 7, 9$) are the coefficients of the harmonics, α and β refer to amplitude of the signals, and the remaining harmonics are treated as noise.

Plentiful experimental results prove that the relative content of the harmonics is almost unchanged, which means the proportional relationship between fundamental frequency and the harmonic components is basically unchanged [29]. Thus, the coefficients in (2) can be acquired by experiments according to the FFT of the signals at any speed, and the harmonic coefficients of (2) in our system are obtained in (3) when the fundamental frequency is set to “1”

$$\begin{bmatrix} A_i \\ B_i \end{bmatrix} = \begin{bmatrix} 0.138, 0.0328, 0.0096, 0.0033 \\ 0.138, 0.0328, 0.0096, 0.0033 \end{bmatrix}. \quad (3)$$

Although the harmonic coefficients have been obtained in (3), the amplitude of the raw signals [α and β in (2)] is not equal in real time, which will have an adverse effect on the subsequent algorithm. Therefore, the following normalized operation is performed

$$\begin{cases} \theta_r(k) = \text{atan2}[H_s(k), H_c(k)] \\ H_{s-1}(k) = \sin[\theta_r(k)] \\ H_{c-1}(k) = \cos[\theta_r(k)] \end{cases} \quad (4)$$

where $\theta_r(k)$ indicates the raw position, and $H_{s-1}(k)$ and $H_{c-1}(k)$ are the normalized signals that the amplitude of the fundamental frequency component becomes “1,” k denotes discrete time point. Based on the existing research [15], [21], H_{s-1} and H_{c-1} can be given as

$$\begin{cases} H_{s-1}(k) = \sin[\theta(k)] + C_i \sin[i\theta(k)] \\ H_{c-1}(k) = \cos[\theta(k)] + C_i \cos[i\theta(k)] \end{cases} \quad (5)$$

where C_i ($i = 3, 5, 7, 9$) are the new coefficients of normalized signals, and the recognition method of coefficients C_i is the same as A_i in (2). The new normalized signals and its FFT are shown in Fig. 4, and the new coefficients are given as

$$[C_i] = [0.086, 0.084, 0.0075, 0.014]. \quad (6)$$

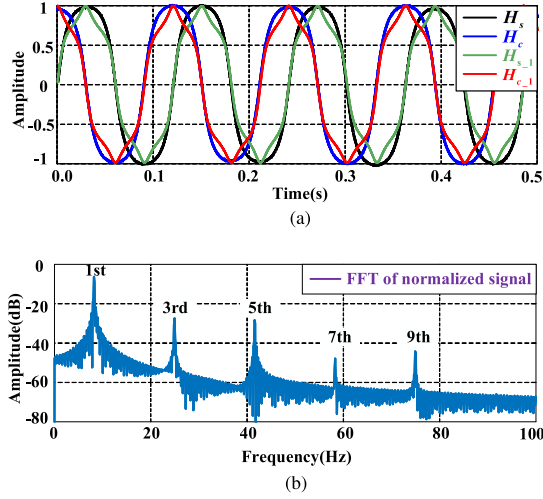


Fig. 4. Normalized information, (a) normalized sinusoidal and cosine signals, (b) FFT of the normalized signal.

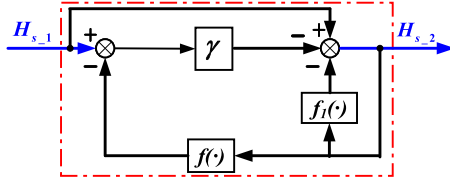


Fig. 5. Diagram of signal reconstruction method.

B. Signal Reconstruction

After the abovementioned normalized operation, the coefficient of the fundamental frequency component becomes “1,” and the coefficients of the harmonics can be approximated as the value shown in (6), which is contributed to the implementation of the subsequent algorithms. According to the new harmonic coefficients, the mathematical models of the signals can be established. Fig. 5 exhibits the signal reconstructed diagram, where the functions $f(\cdot)$ and $f_1(\cdot)$ are the signal models, which are depicted in (8), γ indicates the gain for tuning error. Here, we only take the sinusoidal signal as an example for analysis, and the analysis method of cosine signal is the same as sinusoidal signal.

According to the multiple angle formula, each harmonic can be converted to a function of $\sin[\theta(k)]$

$$\sin[n\theta(k)] = f\{\sin[\theta(k)]\} \quad (7)$$

where n is the order of the harmonic, and $f(\cdot)$ means the function of $\sin[\theta(k)]$. On the basis of (7), the functions $f(\cdot)$ and $f_1(\cdot)$ in Fig. 5 can be given in the following equation:

$$\begin{aligned} f(\cdot) = & H_{s-2}(k) + C_3 [3H_{s-2}(k) - 4H_{s-2}(k)^3] \\ & + C_5 [5H_{s-2}(k) - 20H_{s-2}(k)^3 + 16H_{s-2}(k)^5] \\ & + C_7 [7H_{s-2}(k) - 56H_{s-2}(k)^3 + 112H_{s-2}(k)^5 \\ & - 64H_{s-2}(k)^7] + C_9 [9H_{s-2}(k) - 120H_{s-2}(k)^3 \end{aligned}$$

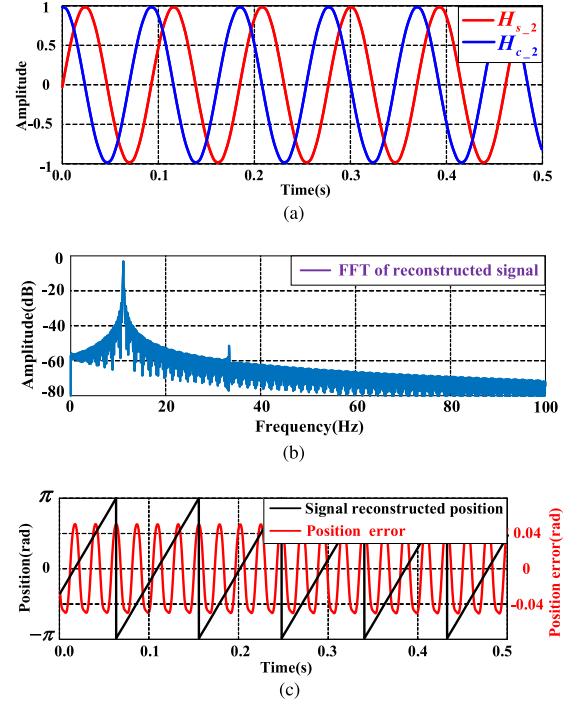


Fig. 6. Signal reconstructed information, (a) reconstructed signals, (b) its FFT, (c) position and position errors.

TABLE I
PARAMETERS OF THIS METHOD

Parameter	Value	Parameter	value
β_1	70	β_2	1230
γ	0.95	h	0.01

$$+ 432H_{s-2}(k)^5 - 576H_{s-2}(k)^7 + 256H_{s-2}(k)^9]$$

$$f_1(\cdot) = f(\cdot) - H_{s-2}(k) \quad (8)$$

where $H_{s-2}(k)$ stands for the output of the signal reconstruction.

After processing by the signal modeling method, the obtained signals, its FFT, the position, and position error are illustrated in Fig. 6. It can be seen that the harmonic contents are greatly reduced, nevertheless, it still contains some noise on account of the coefficient identification error and the unmodeled harmonics, and the position accuracy (maximum error is about 0.051 rad) is still not good enough for the servo motor control, which requires the maximum position error less than 0.013 rad. Therefore, the reconstructed signal still involves some harmonics, and then, H_{s-2} can be depicted as

$$H_{s-2}(k) = \sin[\theta(k)] + G[\theta(k)] \quad (9)$$

where $G(\theta)$ indicates the unconsidered harmonics and modeling errors in (8).

C. Discrete ESO

As mentioned above, the system noise and some harmonics are not considered during signal reconstruction, meanwhile, the modeling errors are also nonnegligible. Thus, the final angular position accuracy still cannot meet the requirements. In order to

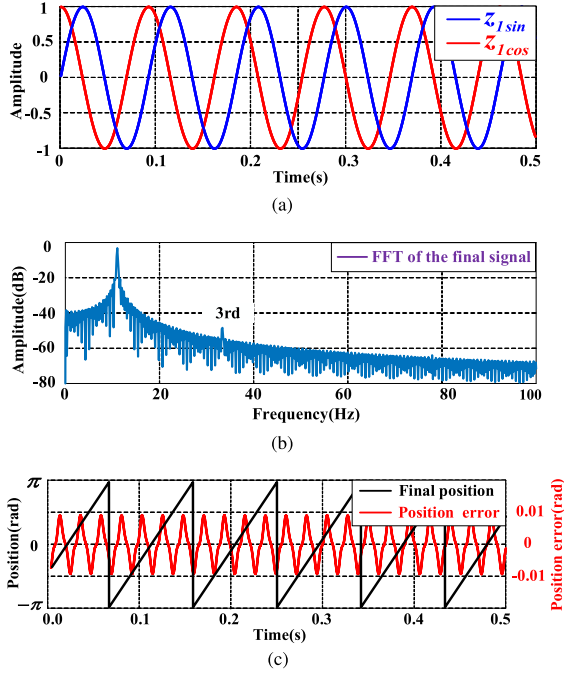


Fig. 7. Final information, (a) final signals, (b) its FFT, (c) final position and position errors.

further improve the position accuracy, the discrete ESO is employed to suppress the influence of the residual noise, which has the advantages of fast convergence speed and high convergence precision. On the basis of (9), the state equation can be set as

$$\begin{cases} \dot{x}_1(k) = u(k) - x_2(k) \\ \dot{x}_2(k) = d(k) \\ y(k) = x_1(k) \end{cases} \quad (10)$$

where $u(k)$ is the output of signal reconstruction $H_{s,2}(k)$, which is regarded as the input of the ESO, $x_1(k)$ means the desired signal $\sin[\theta(k)]$, and $x_2(k)$ refers to the residual noise $G[\theta(k)]$, which is named lumped disturbance, $d(k)$ stands for the acceleration of lumped disturbance, $y(k)$ represents the output variable, which is given as

$$y(k) = -\frac{\cos[\theta_n(k)]}{\omega_n(k)} \quad (11)$$

where $\theta_n(k)$ stands for the final angular position, which will be given later, and $\omega_n(k)$ indicates the final speed, which can be regarded as a constant in a sampling period in the low-speed servo system.

Consequently, the discrete observer can be depicted as

$$\begin{cases} e(k) = z_1(k) - y(k) \\ z_1(k+1) = z_1(k) + h[z_2(k) - \beta_1 e(k) + u(k)] \\ z_2(k+1) = z_2(k) + h[-\beta_2 e(k)] \end{cases} \quad (12)$$

where $e(k)$ indicates the observation error, $z_1(k)$ and $z_2(k)$ stand for the estimation of $x_1(k)$ and $x_2(k)$, respectively, and h indicates the integration step.

However, a new problem comes out that the average value of the results is not zero, which will be different at different speed

TABLE II
SMSCMG PARAMETERS

VARIABLE	VALUE
Maximum angular speed of gimbal system	15 deg/s
Harmonic reducer reduction ratio	100:1
Number of motor pole pairs	8

when calculating the last two formulas in (12), and this will cause new dc component to the two signals and new position errors to the angular position. In order to handle this problem, the following discrete integral method is adopted to process the signal $z_2(k)$

$$z_2(k+1) = z_2(k) + h\{-\beta_2 e(k) - z_2(k-m)\} \quad (13)$$

where m represents the number of integration points. According to (13), the single-step method is adopted ($m=1$), and (12) can be rewritten as

$$\begin{cases} e(k) = z_1(k) - y(k) \\ z_1(k+1) = z_1(k) + h[z_2(k) - \beta_1 e(k) + u(k) - z_1(k-1)] \\ z_2(k+1) = z_2(k) + h[-\beta_2 e(k) - z_2(k-1)] \end{cases} \quad (14)$$

Therefore, the final position is expressed as

$$\begin{cases} \theta_n(k) = \text{atan2}(z_{1\sin}(k), z_{1\cos}(k)) \\ \omega_n(k) = [\theta_n(k) - \theta_n(k-1)]/T \end{cases} \quad (15)$$

where $z_{1\sin}(k)$ and $z_{1\cos}(k)$ stand for the output variables of the discrete ESO, $\theta_n(k)$ indicates the final desired angular position, $\omega_n(k)$ is the angular speed which is utilized in (11), T refers to the sampling time. Fig. 7 exhibits the the final signals, its FFT, the final position, and the position errors.

It can be seen from Fig. 7(b) that the harmonic contents of the signals are greatly decreased. Besides, there is almost no phase lag in the final position, and the angular position error in Fig. 7(c) is less than 0.01 rad, which is acceptable for the gimbal servo control.

The abovementioned analysis is the details of the method proposed in this article for the problem depicted in Section II, and the diagram of the proposed SEM for extracting desired position is shown in Fig. 8.

IV. SIMULATION

To demonstrate the ability of the proposed method, several simulations are performed using MATLAB/SIMULINK software. The parameters of this method are given in Table I.

On the basis of parameters in Table I and the diagram in Fig. 8, the simulation model can be established. In order to demonstrate the effectiveness of the proposed method while avoiding verbosity, only the FFT and the position errors of the raw signals and final signals are exhibited in Fig. 9 at three different gimbal angular speeds (ω_g), which indicate the low speed, medium speed, and high speed of the gimbal servo system, respectively.

It can be seen from Fig. 9 that the harmonics in the final signals are almost suppressed with only a few third harmonics (maximum 0.3%) left at any speed. Consequently, the accuracy of the final positions (maximum error of about 0.01 rad) is

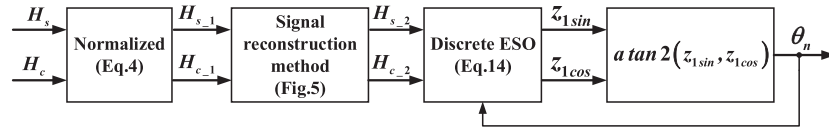


Fig. 8. Diagram of the proposed SEM.

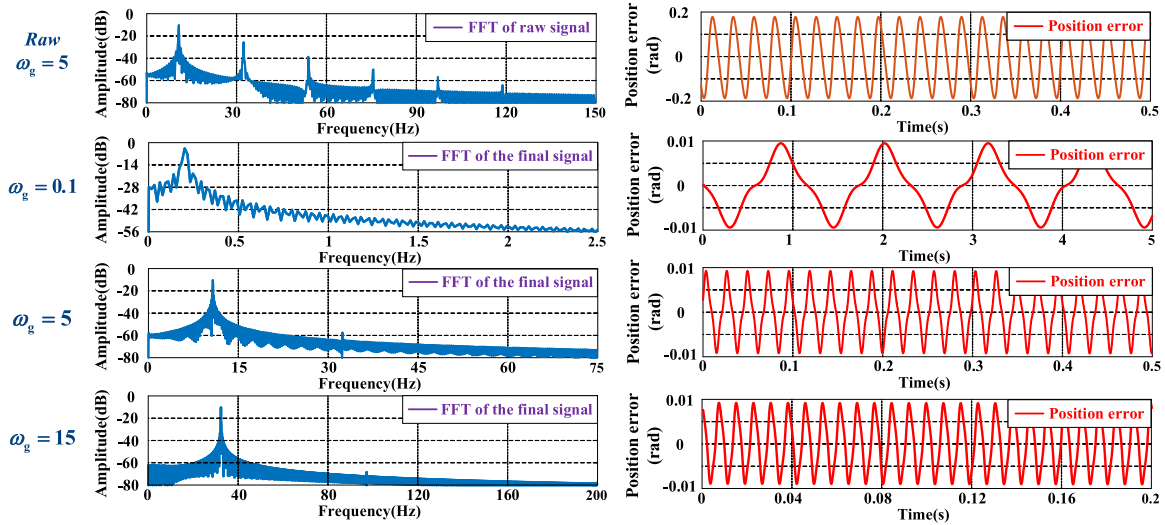


Fig. 9. FFT of raw and final signals and position errors at three different gimbal angular speed.

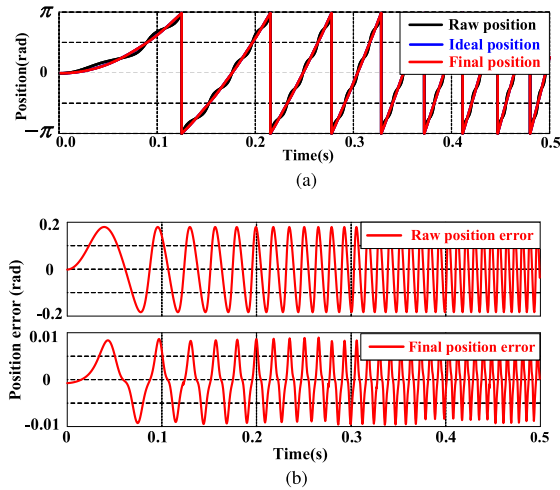


Fig. 10. Position information at time-varying speed, (a) positions obtained by different methods, (b) final position errors.

improved a lot compared to the raw position accuracy (position error of about 0.18 rad). Apparently, the final position fully meets the requirements of the servo system's position accuracy for high-precision control.

As for the gimbal system in SGMSCMG, it often works at the state of time-varying speed motion. Therefore, Fig. 10 illustrates the angular position and the position error in the state of uniform acceleration of about $1500^\circ/\text{s}^2$, which is much larger than the acceleration of the actual system (less than $400^\circ/\text{s}^2$).

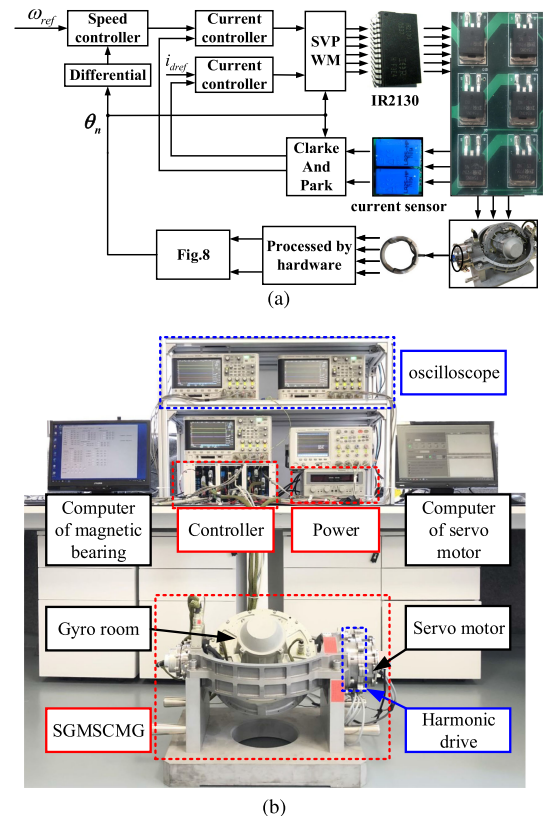


Fig. 11. Experiments of (a) diagram of the control scheme, (b) experimental setup.

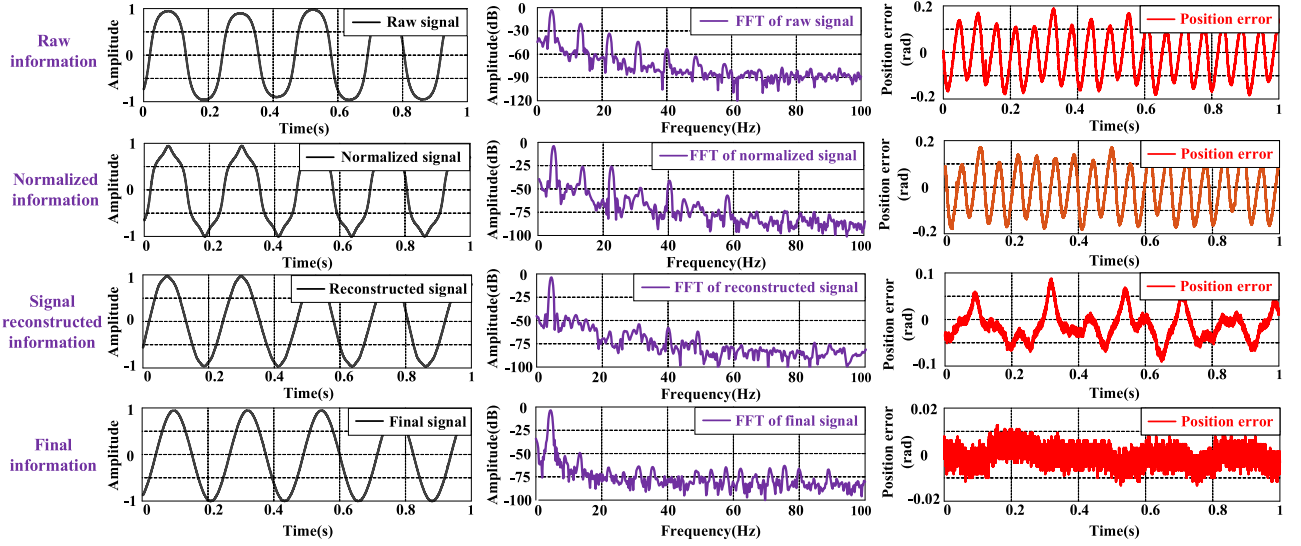


Fig. 12. Raw, normalized, signal reconstructed and final signals, FFT, and the position errors.

V. EXPERIMENTS

In order to verify the effectiveness of the proposed method, according to the simulations, the experiments are carried out on an SGMSCMG, and the main parameters of the gimbal system are shown in Table II.

There are two control chips used in the experiments, one is the TMS320F28335 (DSP), the other is A3P250-2PQ208I (field-programmable gate array). The control scheme and the experimental setup are depicted in Fig. 11

Fig. 12 exhibits the raw signal, normalized signal, reconstructed signal, the final signal, their FFT, and the position errors. It can be seen from the second column of Fig. 12 that the harmonic components are almost decreased in the final signal. The final position errors are also suppressed from about 0.18 rad to about 0.011 rad, which means the final position can satisfy the control requirements obviously. Numerous experiments show that the final angular position errors can maintain within 0.013 rad under the state of uniform speed, and it implies that the final position can fulfill the control accuracy requirement for the servo control system.

When the servo motor operates at low and time-varying speed, the position and position errors are depicted in Fig. 13, and it is at a uniform acceleration state with an acceleration of $400^\circ/\text{s}^2$, which is approximately the maximum acceleration of the gimbal servo system.

From Fig. 13, we can get that at uniform acceleration state, the proposed method still remains its effectiveness, and the position error becomes about 0.012 rad. Combining with the results in Fig. 12, it can be seen that the proposed method can meet the position accuracy requirements at any state of the gimbal system. Furthermore, the results of repeated experiments confirm that the method can still purify position information, and the position accuracy requirements of the control algorithm are also satisfied.

In the aforementioned experiments, the position information of different methods at both constant and time-varying speed is

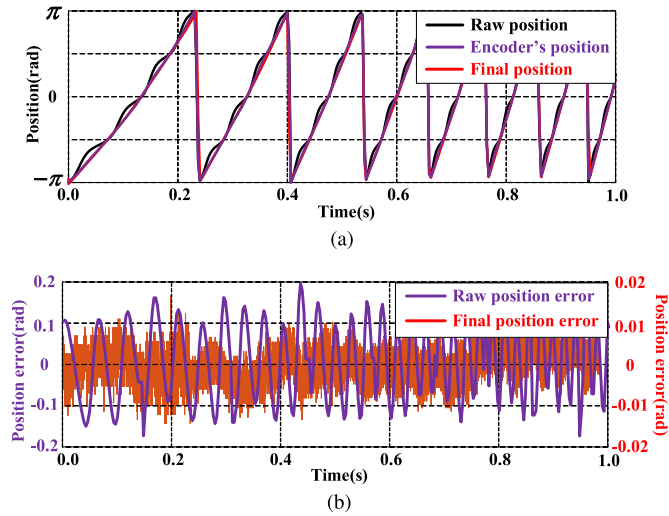


Fig. 13. Position information at time-varying speed of (a) different methods' positions, (b) position errors.

It can be seen from Fig. 10 that even if the acceleration is large, the angular position accuracy can still be kept within 0.013 rad, which is similar to the position accuracy at constant speed. Therefore, we can know from the results in Figs. 9 and 10 that the suppressed effect of the proposed method on the disturbance in the signals is basically independent of speed, and the effectiveness of the proposed SEM can be demonstrated according to the abovementioned simulation.

In summary, plentiful simulations validate that the proposed method has a good performance of extracting the raw signals with abundant harmonics and unequal amplitude in both steady state and dynamic state, and the final angular position accuracy can satisfy control requirements.

exhibited, and results prove that the final position can all meet the position accuracy requirements of the gimbal servo system. Consequently, the effectiveness of the proposed SEM in this article is demonstrated.

VI. CONCLUSION

In this article, the problem of extracting the desired angular position from the poor signals detected by the linear Hall sensors was studied in the ultralow and time-varying speed system. According to the existing research results, a new composite SEM was proposed for the special problems in the gimbal servo system with the following three advantages.

- 1) The accurate angular position information can be acquired without knowing the fundamental frequency information in advance.
- 2) This method can be applied in low speed and medium speed conditions, and the accurate position information can be obtained.
- 3) The method is simple and practical, and it is conducive to the engineering application.

Finally, the effectiveness of this method was verified by plentiful simulation and experiments in both steady state and dynamic state, and the position accuracy can meet the requirements of the gimbal servo system in SGMSCMG.

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